LESSON 9.4

Apply Two-Variable Relationships to Solve Problems

Set B · Version A · Second form · supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write an equation to represent the relationship between two variable quantities and use it to solve a real problem.

WHY THIS MATTERS An automobile plant's assembly line runs in 8-hour shifts and produces 2.5 cars per hour. An order arrives: 1,000 cars in 3 weeks. Should the manufacturer accept? Later, the student council plans a fundraiser car wash — last year they washed 34 cars and raised $408, and this year the goal is $500.

REMEMBER

Apply Two-Variable Relationships to Solve Problems

Analyzing & Applying Two-Variable Relationships Writing the equation is only half the job. Applying it means substituting what you know, solving for what you don't, and checking whether the answer makes sense in the situation.

1. Connect the four representations: Scenario, Table, Graph, and Equation.
2. Substitute given values into the equation to make predictions and solve problems.
3. Interpret coordinates and rate of change in the real-world context

Words: solution *(solución)* · substitute *(sustituir)* · at least *(al menos)* · predict *(predecir)* · justify *(justificar)*

PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each statement about c = 2.5h.

Groups: True · False

_____ In 4 hours the line produces 10 cars

_____ h is the independent variable

_____ The unit rate is 2.5 cars per hour

_____ In 2 hours the line produces 4 cars

_____ c is the input of the equation

_____ In 5 hours the line produces 25 cars

HINT Test the number claims by substituting into c = 2.5h.

2 **Complete the table. Show how you found each value.**

The assembly line produces 2.5 cars per hour. Complete the production table.

Hours (h)	Substitute c = 2.5h	Cars produced (c)
4		
8 (one shift)		
100		

HINT Multiplying by 2.5 is multiplying by 2, plus half of the number.

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A table pairs x = 1, 2, 3 with y = 4, 8, 12. Which equation fits?

(A) y = x + 3

(B) y = 4x

(C) y = x + 4

(D) y = x ÷ 4

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A bike travels 12 miles every hour. Which equation gives distance d after h hours?

(A) d = 12h

(B) d = 12 + h

(C) d = h ÷ 12

(D) d = 12 − h

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

In the equation y = 5x, what is y when x = 6?

(A) 5

(B) 11

(C) 30

(D) 56

My work — one step per line	What I did to both sides

1. **Look** *What is done to the variable?*
2. **Undo** *Do the opposite — both sides.*
3. **Simplify** *One side at a time.*
4. **Check** *Put it back in.*

EXPLAIN YOUR THINKING

One rate is cars per hour and the other is dollars per car. Ask your partner to predict one value from each, then check by substituting — which prediction could you make without a calculator, and why?

SENTENCE STARTER Substituting ___ gives ___, so I predict ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

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1 **Circle the letter of the best answer. Show how you know.**

When you write an equation for a relationship, which variable is usually alone on one side?

(A) The dependent variable
(B) The independent variable
(C) Whichever is larger
(D) It does not matter

SHOW YOUR WORK

2 **Circle the letter of the best answer. Show how you know.**

Yelina is comparing the costs of having cupcakes delivered from two different bakeries. Each bakery has its own relationship between the number of cupcakes ordered and the total cost. Yelina has a budget of $110 to spend. Bakery A charges $2 per cupcake plus a $10 delivery fee, so its cost equation is $c = 2n + 10$, where n is the number of cupcakes. Substituting Yelina's $110 budget, which equation should she solve to find n?

(A) $2n + 10 = 110$
(B) $2n + 10 = 100$
(C) $2n = 110 + 10$
(D) $n = 2(110) + 10$

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Yelina is comparing the costs of having cupcakes delivered from two different bakeries. Each bakery has its own relationship between the number of cupcakes ordered and the total cost. Yelina has a budget of $110 to spend. Solving 2n + 10 = 110 for Bakery A, how many cupcakes can Yelina buy?

(A) 45 cupcakes
(B) 50 cupcakes
(C) 55 cupcakes
(D) 60 cupcakes

My work — one step per line	What I did to both sides

4 **Circle the letter of the best answer. Show how you know.**

Yelina is comparing the costs of having cupcakes delivered from two different bakeries. Each bakery has its own relationship between the number of cupcakes ordered and the total cost. Yelina has a budget of $110 to spend. Bakery B charges $2.50 per cupcake with no fee: cost = 2.5n. Solving 2.5n = 110 gives n = 44 cupcakes. Which bakery should Yelina choose to get more cupcakes within her $110 budget?

(A) Bakery A, because 50 cupcakes is more than 44 cupcakes
(B) Bakery B, because 2.5 is greater than 2
(C) Bakery A, because it has a delivery fee
(D) They are the same, because both cost exactly $110

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

The plant produces 2.5 cars per hour. Using 2.5h = 1,000, how many hours does the order of 1,000 cars take?

(A) 40 hours
(B) 400 hours
(C) 997.5 hours
(D) 2,500 hours

PER 1	

EXPLAIN YOUR THINKING

One rate is cars per hour and the other is dollars per car. Ask your partner to predict one value from each, then check by substituting — which prediction could you make without a calculator, and why?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

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